



CLINICAL PROGRAMMING BOOTCAMP • MODULE 11

Relationships & Trial Design

How datasets point at each other, and where the protocol lives

Two loose ends from earlier: the SUPPQUAL you already built, and the trial-design datasets that describe the study itself rather than its subjects.

Concept module — lighter treatment; no new notebook.

Two mechanisms, two jobs

SUPPQUAL

Attach EXTRA information to a record.

A standard domain has a fixed set of variables. When you must carry a value that has no standard home, it goes in a Supplemental Qualifiers dataset — NOT a new column.

You already built one

SUPPAE, carrying AETRTEM — the treatment-emergent flag that is not a standard AE variable.

SUPP-- links to its parent by
USUBJID + RDOMAIN + IDVAR + IDVARVAL

RELREC

Relate one RECORD to another RECORD.

When two records in DIFFERENT domains describe the same clinical event, RELREC records the link — formally, so a reviewer does not have to infer it.

The ABC-01 example you have seen

The AE "bad headache" and the CM "Paracetamol" taken for it — event and treatment, one clinical story, two domains.

RELREC links by
USUBJID + RDOMAIN + IDVAR + IDVARVAL + RELID

SUPPQUAL adds a QUALIFIER to a record; RELREC connects a RECORD to another record. Different jobs, similar machinery.

Every SUPP-- dataset has the same eight-ish columns

Variable	Holds	In SUPPAE
STUDYID	the study	ABC-01
RDOMAIN	which domain this qualifies	AE
USUBJID	which subject	ABC-01-01-002
IDVAR	how we point at the parent	AESEQ
IDVARVAL	which record specifically	1
QNAM	the qualifier's name	AETRTEM
QLABEL	its human-readable label	Treatment Emergent Flag
QVAL	the value	N
QORIG / QEVAL	where it came from / who assessed	DERIVED / (blank)

IDVAR + IDVARVAL is the key. IDVAR names the parent variable (AESEQ), IDVARVAL gives its value (1). Together they say "this qualifier belongs to AE record AESEQ=1 for this subject" — which is exactly the link an unstable --SEQ sort would break.

Why not just add AETRTEM as a column in AE?

If you added a column...

AE would no longer match the standard AE domain. A validation tool compares your AE against the published SDTMIG structure and flags every non-standard variable as an ERROR.

Worse, a reviewer's tools expect AE to have exactly the standard columns. An extra one breaks the automation they rely on to read fifty submissions the same way.

So SUPPQUAL exists

The standard domains stay EXACTLY as published — same columns, every study, every sponsor. Anything extra goes in a SUPP-- dataset with a fixed generic shape.

The cost is that a qualifier lives in a separate dataset and must be merged back. That is a price worth paying for domains that are identical across the entire industry.

The rule, one more time: never add a non-standard variable to a standard domain.

Trial Design describes the STUDY, not the subjects

Everything you have built is subject data. Trial Design is not.

DM, AE, VS and the rest have one thing in common: a row is about a SUBJECT. The Trial Design datasets describe the PROTOCOL itself — the planned arms, the planned visits, the study's parameters. There is no USUBJID, because these are facts about the study, true before a single subject enrolled.

TA	Trial Arms	the planned treatment arms and the sequence of elements in each
TE	Trial Elements	the building-block periods an arm is made of (Screen, Treatment, Follow-up)
TV	Trial Visits	the planned visit schedule — the source of VISITNUM
TS	Trial Summary	one-row-per-parameter facts about the study: phase, indication, objectives

What TS would say about ABC-01

TSPARMCD	TSPARM	TSVAL
STUDYID	Study Identifier	ABC-01
TITLE	Trial Title	A Phase 2 Study of Drug A vs Placebo
PHASE	Trial Phase	PHASE II TRIAL
TTYPE	Trial Type	TREATMENT
TCNTRL	Control Type	PLACEBO
INDIC	Indication	(the condition under study)
ACTSUB	Actual Number of Subjects	8

One row per parameter — a vertical structure again

TS is the same tall shape as a Findings domain: one fact per row, identified by a code (TSPARMCD) from a controlled list. It is how a reviewer — or an automated system — reads the study's headline facts without opening the protocol PDF. ABC-01 does not ship a TS in this bootcamp, but every real submission does.

Values shown are illustrative of the ABC-01 design; TSPARMCD values come from the CDISC Trial Summary codelist.

Where everything you have built sits

Special Purpose	DM	one row per subject — demographics and reference dates
Interventions	EX · CM	what was given to the subject
Events	AE · DS	what happened to the subject
Findings	VS · LB	what was measured
Relationships	SUPPAE · (RELREC)	extra qualifiers and record links
Trial Design	TA · TE · TV · TS	the protocol itself — no subjects

Five of the six you have built end to end. Trial Design and RELREC are the pieces this module named — you now know the whole SDTM package, not just the domains with subjects in them.